

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1093

Kinsington 100 kV
February 22, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1093. The Interconnection Customer proposes a 193.0 MW wind facility interconnecting at the Kinsington 100 kV substation in the Illinois area. The request was received on October 30, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 390 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,038,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kinsington to River Rock 100.0 kV	Jacobs Corner to Kinsington Substation 100.0 kV Line	122.2	140	100.0
Jacobs Corner to Kinsington 100.0 kV L	River Rock to Kinsington 100.0 kV line	120.6	142	100.0
Imperial to Chamberlin 230.0 kV	Hylebos to Ravenna Substation 500.0 kV	101.4	1,691	10.6
Canal Rd to Weedpatch 230.0 kV Line	Trego to Beaver 500.0 kV Line	101.4	1,546	7.5
Twin County Ranch to Swan Vly 230.0 kV	Richland to Troy Energy Llc 500.0 kV	100.5	1,145	17.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Riv Rock to Kinsington 100.0 kV	Rebuild line with higher-capacity conductor	\$4,539,000	100% (\$4,539,000)
Jacobs Corner to Kinsington 100.0 kV	Rebuild line with higher-capacity conductor	\$20,048,000	100% (\$20,048,000)
Imperial to Chamberlin 230.0 kV	Reconductor line	\$7,519,000	2% (\$112,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Canal Rd to Weedpatch 230.0 kV	Reconductor line	\$7,507,000	2% (\$112,000)
Twin County Ranch to Swan Valley 230.0 kV	Reconductor line	\$15,148,000	2% (\$227,000)
	Total allocated to project		\$25,038,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.